Green Energy and Intelligent Transportation 5 (2026) 100293

Contents lists available at ScienceDirect

Green Energy and Intelligent Transportation

journal homepage: www.journals.elsevier.com/green-energy-and-intelligent-transportation

Review article

Smart parking systems: A comprehensive review of digitalization of parking services

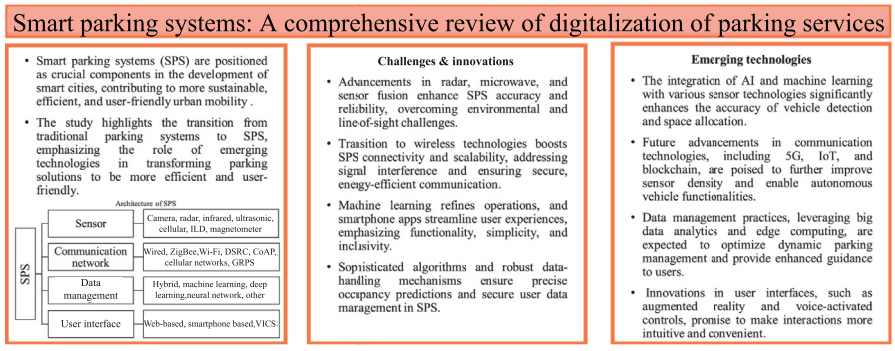
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HIGHLIGHTS

- Significant research growth in smart parking systems (SPS) observed from 2016 to 2023, driven by AI and IoT advancements.
- Camera and radar sensors are prominently used for vehicle detection in SPS.
- Wireless communication technologies, such as ZigBee and Wi-Fi, are increasingly favored over wired options.
- Hybrid prediction models and machine learning techniques enhance SPS accuracy and data analysis.
- Mobile apps dominate user interfaces for SPS, reflecting the rise in smartphone usage.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Smart parking system
Internet of things
Sensor network
Parking occupancy prediction
e-parking

ABSTRACT

Smart parking systems (SPS) address issues that plague traditional parking methods by offering data on real-time parking availability, optimizing the use of space, and facilitating convenient payment solutions. Despite the timeliness and importance of the systems, however, the literature fails to adequately identify areas within SPS that can be vastly improved by innovation. This study addresses the research gap by identifying the key limitations of 124 comprehensively reviewed academic papers and offering innovative solutions. For sensor technology, the challenge of environmental effects and camera line-of-sight issues is tackled with a proposed integrated sensor framework, combining radar precision with camera coverage, all enhanced by AI for greater detection accuracy. Communication networks, currently hindered by scalability and node failure, could be improved with a mesh network architecture for better reliability. To address data management concerns, specifically data integrity and security, the integration of blockchain technology is suggested to protect against data breaches and boost user confidence. Lastly, to simplify complex SPS user interfaces, AI-driven adaptive interfaces are recommended to personalize the user experience and improve system engagement. The findings of this study will be instrumental for city planners, SPS developers, and parking authorities who are tasked with implementing efficient and reliable smart parking solutions.

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https://doi.org/10.1016/j.geits.2025.100293
Received 13 May 2024; Received in revised form 8 October 2024; Accepted 13 October 2024
Available online 26 March 2025
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1. Introduction

Locating a parking spot may seem straightforward, but it's actually a complex task demanding significant amounts of time, patience, and attention from drivers. Studies show that in areas with heavy traffic, 30 to 50 percent of drivers use about 3.5–14 min to find parking [1,2]. This prolonged search contributes to frustration among drivers, exacerbates traffic jams, and leads to higher levels of pollution. However, when drivers have access to up-to-date information about parking spaces, they're 45% more efficient in parking compared to those without such data [3], which also helps in reducing the environmental impact of emissions from vehicles that are otherwise in motion looking for parking [4].

Over recent years, technological advancements have profoundly altered the global landscape. As these technologies progress, tools like the Internet of Things (IoT), machine learning, AI, and big data are becoming integral to our daily existence and are set to redefine our entire living environments. The parking sector is a prime example of this transformation, with digitalization leading the charge in developing smart urban environments. Traditional parking facilities are increasingly seen as inefficient, unable to meet the rising demands from urban growth and traffic increases. This recognition is fueling the push towards more intelligent parking solutions. Research by Huang et al. (2022) suggests that these developments are central to the evolution of smart cities [5]. A report by Research and Markets forecasts that from 2022 to 2030, the global market for smart parking will see a significant growth rate of 18.47% annually, potentially reaching a market worth of USD 21 billion by the end of this period. This growth trajectory marks a substantial uptick from a USD 5.3 billion valuation in 2022, indicating the sector's robust expansion [6].

Smart parking systems (SPS) have seen substantial evolution in response to the complexities that come with managing parking spaces. The advent of parking meters and ticketing systems in the 1930s marked a pivotal point by introducing a regulated framework for timed parking and fee collection [7], while the 1980s witnessed the advent of electronic parking meters, further refining the process of automated fee collection and law enforcement [8]. As sensor technologies emerged at the close of the 20th century, they gave rise to sensor-based parking systems which, according to Bhagwat et al. (2007), leverage real-time data on parking space availability. This information is transmitted to drivers through the integration of wireless technologies, including Wi-Fi and cellular networks [9]. Mobile applications have also been created, offering details on parking availability, booking, and navigation, with the incorporation of IoT-based frameworks enabling smooth communication between sensors, devices, and infrastructure, thereby enhancing data management [10]. The adoption of data analytics and AI in SPS has been noted to increase, providing advanced parking management, predictive analytics, and driver-centric personalized services [11].

SPS play a crucial role in promoting green transportation by reducing vehicle emissions, improving traffic flow, and supporting alternative transportation modes. By providing real-time data on parking availability, SPS minimizes the time drivers spend searching for parking, thereby decreasing idle time and associated emissions. For instance, the implementation of SPS in San Francisco has led to a reduction in CO_2 emissions by 10%–30% [12]. Furthermore, SPS helps streamline traffic, reducing congestion and its environmental impact, as seen in Barcelona, where smart parking systems have contributed to a 21% reduction in traffic congestion [13]. Additionally, SPS often include features that promote the use of electric vehicles (EVs) and car-sharing programs, supporting the transition to greener transportation options. In Oslo, the integration of EV charging stations within the SPS framework has increased the adoption of EVs by 15% [14]. Efficient land use is another significant benefit, with cities like Tokyo repurposing former parking lots into green parks, enhancing urban green spaces [15]. The data generated by SPS also aids urban planners in making informed decisions about sustainable urban development and transportation policies. New York City leverages SPS data to design bike lanes, pedestrian zones, and efficient public transit routes, fostering a comprehensive approach to green transportation [16].

The myriad advantages of SPS have significantly attracted research efforts aimed at addressing its inherent challenges and advancing innovative strategies. However, the existing literature showcases a series of studies and reviews which, despite their informative nature, exhibit certain shortcomings. A detailed comparison of these studies is presented in Table 1, highlighting the gaps and limitations that this review aims to address. For instance, studies such as those by Chandrahasan et al. (2016) and Revathi & Dhulipala (2012) offer preliminary surveys on reservation techniques and vehicular sensors, respectively, but lack thorough exploration [17,18]. Similarly, Idris et al. (2009) focused on vehicular detection sensors, providing detailed examples and justifications, yet failed to address other critical SPS components [19]. Reviews by Faheem et al. (2013) and Hassoune et al. (2016) on smart parking methods and software systems provided incomplete hardware details [20,21]. Additionally, Fraifer & Fernstrom (2016) and Al-Turjman & Malekloo (2019) discussed technological advancements without delving into data management systems and user interface design [22,23]. Lin et al. (2017) centered their study on the progression of SPS, yet they simply outlined methodologies and objectives without in-depth analysis [24]. In a comprehensive review, Kaur & Malhotra (2018) examined various SPS but stopped short of analyzing the system's architecture thoroughly [25]. Lastly, Barriga et al. (2019) identified and discussed frequently used components in SPS and observed usage trends but lacked deeper insights into novel solutions [26]. These gaps highlight the need for a more comprehensive and in-depth analysis, which this review aims to address by offering novel insights and solutions for advancing SPS.

The lack of studies that focus on potential areas for innovation in SPS highlights a critical gap in the literature and underscores a need for future research. This study bridges that gap by delivering a thorough evaluation of how cutting-edge technologies is being incorporated into SPS, including sensor technologies, communication networks, data management, and user interfaces, while also highlighting their limitations by critically examining the methodologies employed by the researchers. The objectives of this study were to: (1) Explore the integration of advanced technologies within SPS; (2) analyze diverse methodologies, identifying their strengths and limitations; and (3) identify potential areas for innovation. The outcomes of this study are intended to serve as a robust resource for both researchers and

Table 1
Comparison of existing SPS review studies.

Study	Focus area	Key findings	Gaps/Limitations
[17]	Reservation techniques	Overview of reservation techniques	Lacks thorough exploration and did not address other critical SPS components
[18]	Vehicular sensors	Overview of vehicular sensors	
[19]	Vehicular detection sensors	Detailed examples and justifications of vehicular detection sensors	
[20, 21]	Smart parking methods	Overview of smart parking methods	Incomplete hardware details
[22, 23]	Technological advancements	Overview of technological advancements in SPS	Did not delve into data management systems and user interface design
[24]	Progression of SPS	Overview of SPS methodologies and objectives	Lacked in-depth analysis
[25]	Various SPS	Examined various SPS	Stopped short of analyzing system architecture thoroughly
[26]	Frequently used components in SPS	Identified and discussed frequently used components and usage trends	Lacked deeper insights into novel solutions

practitioners in the smart parking domain, providing a knowledge base that can inform and support future decision-making processes in the development of SPS.

2. Methodology

Fig. 1 illustrates the research methodology utilized in this study to become acquainted with the current body of work on this topic. We began by conducting a comprehensive literature search across a variety of databases, including Google Scholar, Springer, Web of Science, Science Direct, Scopus, IEEE Xplore, and ProQuest. Subsequently, we then employed a range of search terms such as "smart parking system," "Internet of Things," "sensor network," "parking occupancy prediction," and "e-parking" to collate materials from journals, technical articles and conference proceedings published post-2005 that dealt with developing an SPS.

A careful review of the initial 487 documents yielded 198 that were subject to peer review. The selections were further screened to include only the 124 studies that were pertinent to the creation of a SPS. This comprehensive analysis included various methodologies, sensors, prediction models, and e-parking systems covered in these studies, yielding a detailed understanding of the current state of smart parking solutions and the advancements achieved in this area. Through this analytical process, we were able to identify the sector's key strengths and shortcomings, and pinpoint areas requiring additional development.

2.1. Year of publication

Fig. 2 illustrates the increasing interest in the field by depicting the trend in the publication of research articles on SPS from 2006 to 2023. From 2006 to 2009, the number of publications on SPS was relatively low, which may be attributed to the nascent stage of SPS technology and its new emergence into a field of academic inquiry. As technology gained traction and the importance of efficient parking solutions became more evident in urban planning, however, there was a modest increase in research output in the following years, up to 2013. A more significant rise in publications began in 2014, concurrent with the proliferation of IoT and advancements in communication technologies that contributed to new research opportunities and applications for SPS. The substantial jump in 2016–2017 may reflect breakthroughs in related fields such as AI and machine learning, which have direct implications for the efficiency and intelligence of SPS. The strongest surge in research activity was from 2022 to 2023, indicating a peak in interest and possibly the maturity of the field. This recent spike could be due to several factors, including increased urbanization, the push for smart cities, and advancements in autonomous vehicle technologies, all of which present new challenges

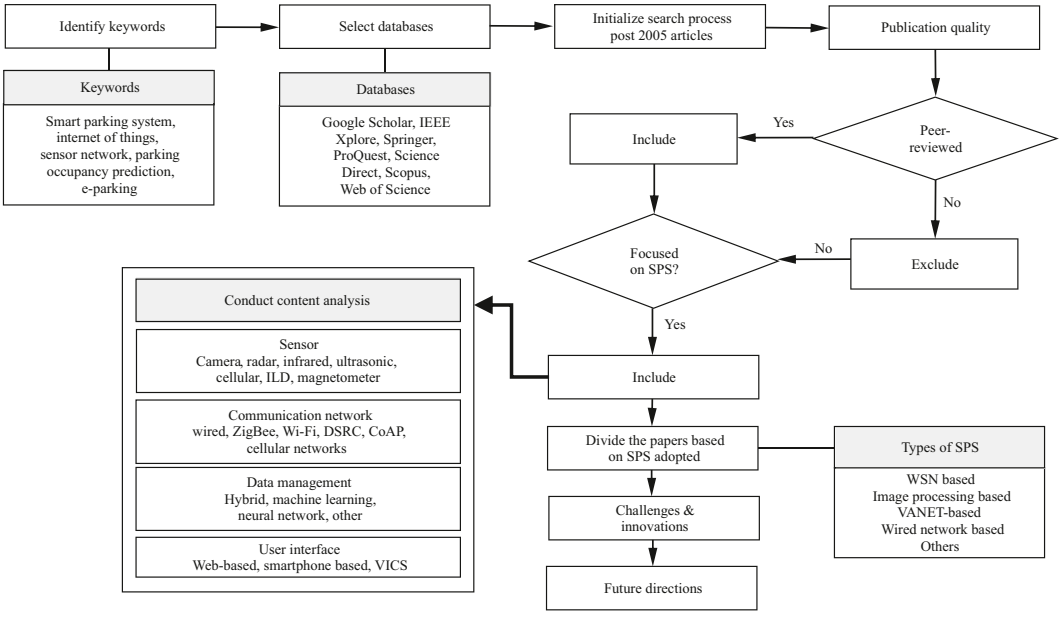


Fig. 1. Research methodology.

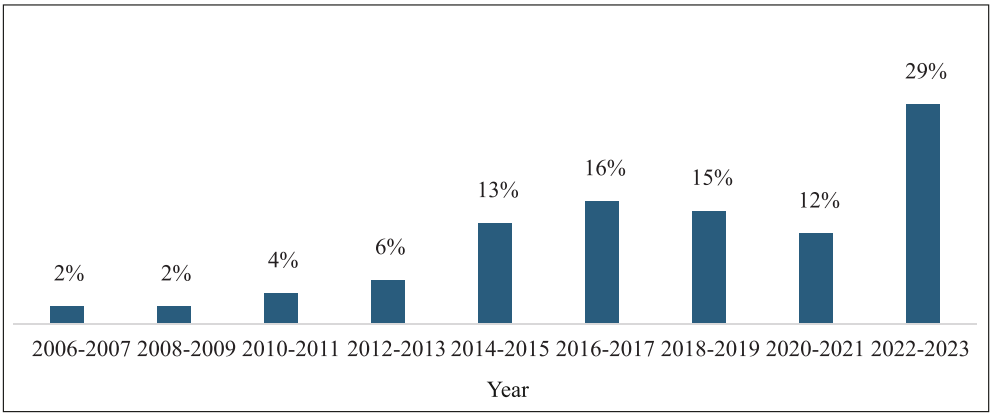


Fig. 2. Research articles published on SPS from 2006 to 2023.

and opportunities for SPS. The overall trend depicted suggests that SPS has evolved from an emerging area of research to a well-established field, with its importance continuing to grow as cities worldwide seek smarter and more sustainable traffic and parking solutions.

2.2. Overview of SPS structure

SPS integrates several critical components to enhance parking management and user experience. The primary elements of an SPS include sensors, communication networks, data management systems, and user interfaces. Sensors such as cameras, ultrasonic sensors, radar, and magnetometers are employed to detect the presence of vehicles in parking spaces, providing real-time data on parking availability. For example, radar sensors have a detection accuracy of 95% even in adverse weather conditions, while camera sensors provide detailed visual information but may have an accuracy drop of up to 10% in low light conditions. Ultrasonic sensors, known for their cost-effectiveness, maintain an accuracy rate of 85% in all weather conditions. Communication networks including Wi-Fi, ZigBee, cellular networks (e.g., GPRS, 3G, 4G, 5G), and dedicated short-range communications (DSRC) facilitate the transmission of data between sensors, central systems, and user interfaces. For instance, Wi-Fi provides high-speed data transfer rates up to 1 Gbps with a range of 100 m, whereas ZigBee offers low-power communication with a data rate of 250 kbps and a range of up to 10–20 m. Cellular networks, especially 5G, offer data rates up to 10 Gbps and low latency of less than 1 ms, making them suitable for real-time applications in extensive parking areas. Data management systems such as cloud-based platforms, blockchain technology, and edge computing are responsible for processing, storing, and analyzing the vast amounts of data generated by the sensors, ensuring data integrity, security, and efficient handling of real-time and historical data. For example, blockchain technology enhances data security by providing a tamper-proof ledger with an average transaction processing time of 10 min per block, significantly reducing the risk of data breaches. Edge computing reduces latency by processing data locally, with response times improved by up to 50% compared to traditional cloud-based systems. User interfaces, which can be web-based, smartphone-based, or integrated into vehicle information and communication systems (VICS), provide end-users with information about parking availability, booking options, and navigation assistance. This comprehensive structure of SPS components works together to streamline parking operations and improve the overall user experience.

2.3. Research landscape overview of technologies used in SPS

Fig. 3 groups the various technologies utilized in studies related to SPS into categories of sensors, communication networks, prediction models, and user interfaces. As shown, cameras are frequently chosen for vehicle detection in SPS sensor studies, while radar and infrared sensors are favored for their all-weather robustness. Ultrasonic and cellular technologies are also key for their proximity detection and wide-area connectivity, respectively. Less common are inductive loop detectors (ILDs) and magnetometers, which are used for niche applications like distance measurement and magnetic detection.

The communication network shift from wired to wireless technologies, emphasizing low-power ZigBee for sensor networks and high-speed Wi-Fi for data transfer, is explored for its unique contributions to vehicle communication and remote activity, as are DSRC, CoAP and GPRS. Prediction model research favors hybrid methods, as they combine computational models for better accuracy. Machine learning is valued for pattern recognition in large datasets, and neural networks are being investigated for their advanced data interpretation capabilities. User interface studies predominantly focus on smartphone apps, reflecting the surge in mobile usage and the convenience of managing parking-on-the-go. Web-based solutions remain significant, suggesting their continued relevance, while VICS is considered for its potential integration with in-vehicle systems.

3. Architecture of SPS

An SPS integrates several elements that collectively enhance parking administration and elevate the user's experience. It involves the deployment of sensors in parking spots to monitor the presence of vehicles, a communication infrastructure to facilitate the transfer of data, a data management system dedicated to handling and analyzing the data, and an application for users that provides up-to-date information and booking capabilities [18].

3.1. Sensors

Researchers have been innovative in designing smart parking systems that can achieve accurate and efficient parking management and in their selection of devices that can detect a vehicle in a space. Ma et al. (2017) used ultrasonic sensors to measure distances and detect the presence of

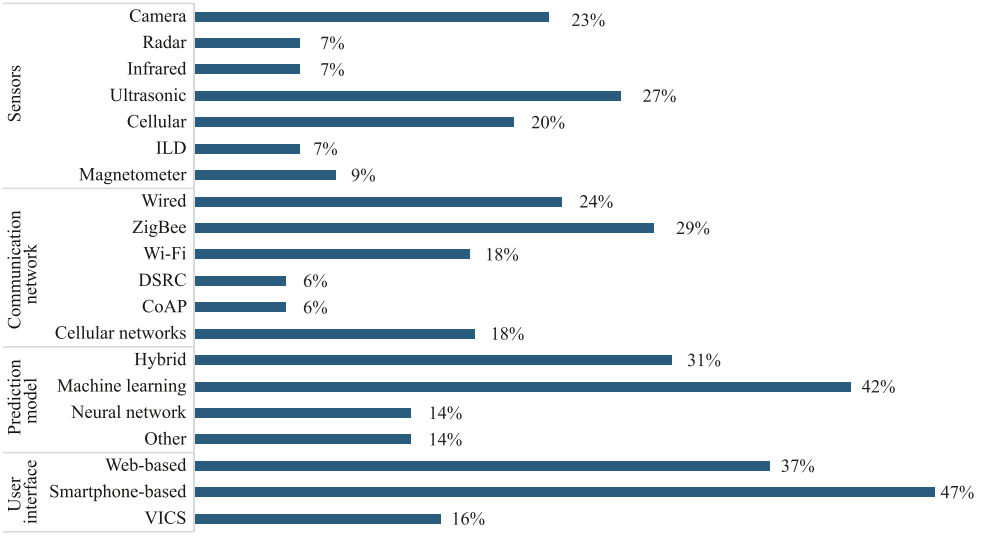


Fig. 3. Distribution of technologies utilized in SPS by different studies.

vehicles [27]; Garcia et al. (2018) utilized radar sensors' sound waves for the same purpose [28]. Gao et al. (2018) employed cellular sensors to leverage cellular network connectivity to detect the presence of vehicles [29], and Baroffio et al. (2016) used camera sensors to process images and analyze vehicles in parking areas [30]. Amato et al. (2016) employed an electromagnetic field generated by a loop of wire embedded in pavement to reveal vehicles [31], and Ionita et al. (2018) employed magnetometers that detected changes in the magnetic field [32]. Table 2 presents a summary of the various types of sensors employed by researchers.

The use of various types of sensors illustrates the breadth of innovation within the field of SPS. Each type brings unique advantages and capabilities to the table, from ultrasonic sensors' cost-effectiveness and reliability in adverse weather conditions [41,47] to the precision and real-time operation of radar sensors [28]. Camera sensors represent a significant innovation by providing detailed visual information that not only aids in vehicle detection but also facilitates advanced features like license plate recognition [30]. The application of electromagnetic fields and magnetometers introduces a nuanced approach to vehicle detection by leveraging changes in the magnetic field or electromagnetic signatures and offers a novel sensor technology application in parking management [31,32].

3.2. Communication network

Wired or wireless sensor communication networks serve as a link between sensors and a processing unit and enable it to transmit data. Table 3 presents the communication and networking technologies adopted in different studies.

The exploration of both wired and wireless communication networks in SPS underscores the field's commitment to robust and efficient data transmission. Wired networks, as highlighted by Amato et al. (2017), offer reliable and secure communication with high bandwidth, making them suitable for environments where consistent connectivity is crucial [42]. Wireless technologies, however, exhibit unprecedented flexibility and scalability into SPS. Technologies such as ZigBee, Wi-Fi, DSRC, CoAP, cellular networks, and GPRS have been instrumental in facilitating sensor connectivity and data transmission across various SPS implementations [19,51,53]. Wireless networks have revolutionized SPS by enabling the easy expansion of systems without the physical constraints and installation costs associated with a wired infrastructure, and the use of cellular networks and Wi-Fi provide the broad coverage and high data rates that are essential for transmitting real-time data and processing in expansive parking environments [51,56].

3.3. Data management system

Data management systems in SPS serve several crucial purposes, ensuring the efficient handling of data generated by the sensors. These purposes include real-time data processing, historical data analysis, and integration with user interfaces. Real-time data processing ensures timely updates on parking availability, which is essential for guiding drivers to available spots. Historical data analysis helps in predicting future parking trends and optimizing parking space allocation. Integration with user interfaces ensures that processed data is presented in a user-friendly manner, enabling easy access to parking information. Table 4 presents the occupancy prediction models adopted by different studies.

The incorporation of diverse computational models, from SVR and neural networks to deep learning frameworks like LSTM, highlights the innovative approaches to data management in SPS. The hybrid model approach, as demonstrated by Liu et al. (2021), and the application of machine learning and deep learning techniques by researchers like Hasan et al. (2021) and Jin et a (2021), showcase the field's move towards more sophisticated, accurate, and efficient data analysis methods [62,71,84]. These models leverage the extensive data generated by SPS sensors to predict occupancy patterns, thereby enabling dynamic parking management and enhanced user guidance. The utilization of neural network architectures and deep learning techniques represents a forward-thinking

Table 2
Sensors utilized in research studies.

Sensor type	Accuracy	Environment	Cost	Advantages	Disadvantages	Refs.
Infrared	90%	All-weather	Moderate	Works in low-light conditions	Susceptible to interference	[33–35]
Ultrasonic	85%–90%	All-weather	Low	Affordable, reliable	Affected by extreme weather	[27,36,37]
Camera	80%–95%	Depends on lighting	Moderate-high	Detailed visual data, license plate recognition	Affected by low-light conditions	[27,38,39]
Magnetometer	85%	Indoor and outdoor	Low	Cost-effective, easy to install	Sensitive to electromagnetic interference	[32,40,41]
ILD	98%	Indoor, ideal for garages	High	High accuracy, reliable	Expensive installation, requires pavement cutting	[19,31,42]
Radar	95%	All-weather	High	Reliable under all conditions	High cost	[28,43,44]
Cellular	90%	Extensive coverage via network	Moderate	Real-time communication, integrates with parking management	Ongoing network subscription costs	[29,45,46]

Table 3
Communication and networking technologies adopted in different studies.

Network type		Advantages	Disadvantages	Latency	Data transfer rate	Refs.
Wired		High reliability, secure	Expensive installation, less flexibility	<1 ms	Up to 10 Gbps	[31,42,48]
Wireless	ZigBee	Low power consumption, cost-effective	Low data rate, short range	~30–50 ms	250 kbps	[37,49,50]
	Wi-Fi	High-speed data transfer, flexible	Limited coverage, higher power consumption	~1–5 ms	Up to 1 Gbps	[28,51,52]
	DSRC	Fast data transfer, suitable for vehicle-to-vehicle communication	Short range, susceptible to interference	~50 ms	Up to 27 Mbps	[53]
	CoAP	Efficient data transmission, cost-effective	Constrained network, limited capabilities	~100 ms	Up to 10 Mbps	[54]
	Cellular networks	Wide-area coverage, real-time data	High cost, potential privacy concerns	<1 ms (5G)	Up to 10 Gbps (5G)	[55,56]
	GPRS	Cost-effective, available widely	Slower data rates compared to 4G/5G	~500 ms	Up to 100 kbps	[57]

Table 4
Occupancy prediction models adopted by different studies.

Type of model	Techniques from existing literature, with sources
Hybrid	F0A7 Support vector regression (SVR) enhanced with fruit fly optimization algorithm [58]
	F0A7 Autoregressive integrated moving average (ARIMA) - multilayer perceptron (MLP) neural network [59]
	F0A7 Convolutional neural network (CNN) - long-short term memory (LSTM) [60]
	F0A7 Ensemble model combining random forest, gradient boosting, and SVR [61]
	F0A7 Recurrent Neural Network (RNN) –LSTM [62]
Machine learning	F0A7 Support Vector Machine (SVM) [63,64]
	F0A7 Random forest [65,66]
	F0A7 Decision trees, SVM, random forests [67,68]
	F0A7 Random forest, KNN, and gradient boosting [69,70]
	F0A7 Random forest, SVR, and MLP [71]
	F0A7 Decision trees, random forest, gradient boosting, SVR, NN [72–74]
	F0A7 LSTM [75–77]
	F0A7 Stacked gated recurrent unit-LSTM [78]
	F0A7 Improved LSTM algorithm [78,79]
Neural network	F0A7 Weighted neural network [80]
	F0A7 Deep CNN [81,82]
	F0A7 Deep NN models with spatial–temporal feature learning [83]
	F0A7 CNN With RNN [84]
Other	F0A7 Simulation-based approach [85]
	F0A7 Multiple regression analysis [86]
	F0A7 Extreme gradient boosting algorithm and LSTM neural network [87,88]

strategy for extracting meaningful patterns from complex parking data sets. These approaches allow for the capturing of intricate temporal and spatial dependencies within the data, leading to more accurate and reliable occupancy predictions and, consequently, more efficient utilization of parking spaces.

3.4. User interface

A user interface is the primary interaction point between users and the SPS. These interfaces aim to facilitate parking space reservation, real-time availability updates, navigation assistance, and payment processing. Table 5 presents an overview of the types of user interfaces for SPS that researchers have investigated.

The exploration of web-based applications and VICS further illustrates the field's commitment to diversifying user access points to SPS, as the varied interfaces cater to different user preferences and situations, ensuring broader accessibility and flexibility. For instance, web-based interfaces provide a platform-independent solution accessible from any device with internet connectivity, while VICS relays parking information directly into the vehicle's navigation system, offering convenience and real-time updates to drivers. The focus on user-centered design in SPS user interface development is evident in the adaptability and personalization features of these systems. The application created by Geng & Cassandras (2012), which allows users to accept or decline a reserved parking space, exemplifies the importance of giving control back to the users, thereby enhancing their decision-making power and satisfaction [89]. Similarly, the study by Mathew et al. (2014) on user choices in parking space allocation underscores the trend towards more personalized and user driven SPS experiences [43].

4. Smart parking systems

The development of an SPS and its integration with IoT and other advanced technologies are a significant improvement over traditional parking meters that are inconvenient and time-intensive, as they have enabled rapid and reliable payment methods that are much more user friendly [90].

4.1. Wireless sensor network (WSN) based SPS

WSN has been fundamental in the evolution of SPS, facilitating real-time tracking and management of parking occupancy. Tang et al. (2006) introduced an intelligent car park management system that relies on wireless sensors to track and manage lot occupancy. This system improves overall parking management by providing real-time data, thus allowing for better resource allocation and reduced congestion [91]. Chinrungrueng et al. (2007) developed an SPS that employs an optical WSN with a star-based topology to direct users to available spots efficiently. This system ensures effective communication and user guidance, reducing the time drivers spend searching for parking [90]. Yan et al. (2011) explored a secure wireless network for parking where vehicles communicate with parking lots via transceivers equipped with infrared devices to detect available spaces. This innovation enhances the security and reliability of parking systems by preventing unauthorized access and ensuring accurate vehicle detection [92]. Yang et al. (2012) proposed a WSN-based SPS prototype that leverages mobile applications for real-time parking status updates and integrates WSN nodes with web servers for easy user access. This approach demonstrates the potential for mobile technology to enhance user experience in SPS [93]. Patil & Bhonge (2013) refined SPS by combining WSN and RFID technologies, utilizing ZigBee for sensor communication. This integration supports continuous monitoring of parking lot occupancy, demonstrating the trend towards sophisticated, user-centric SPS [94]. Grodi et al. (2016) further employed WSN to create a system focused on detecting vehicle occupancy, highlighting continuous innovation in SPS development and showcasing the effectiveness of WSN in real-time data collection and management [47]. Cynthia et al. (2018) introduced an IoT-based SPS that uses infrared sensors and a cloud platform to inform users about parking lot occupancy through a smartphone app. This study highlights the integration of IoT in modern SPS, making parking management more accessible and user-friendly [51].

4.2. Image processing-based SPS

Image processing technologies have significantly advanced the capabilities of SPS by enabling detailed monitoring and automated recognition. Dharmareddy et al. (2013) introduced an SPS using webcams for intelligent image processing to determine parking lot occupancy. This system processes images via an ARM8 microcontroller and relays the information to a web server, allowing users to view parking statuses online or through an LCD at the parking entrance. It also supports parking reservations via SMS facilitated by a GSM module within the controller, enhancing the convenience and functionality of the system [95]. Bura et al. (2018) proposed a system that uses distributed camera networks alongside fog computing and deep learning algorithms to track vehicles,

Table 5
User interfaces employed by studies.

User interface type	Advantages	Disadvantages	Key features	Quantitative metrics	Refs.
Web-based application	Accessible from any device with internet	May not be optimized for mobile devices	Parking availability, reservations	Load time: ~2–5 s, Accessibility: High	[36,54,55]
Smart phone-based application	Convenient, mobile-friendly, real-time updates	Requires app installation, digital divide	Real-time data, mobile payments	Load time: ~1–3 s, User ratings: 4.5/5	[51,52,54]
VICS	Integrated into vehicle navigation systems	Limited to certain vehicles, complex setup	In-car navigation, real-time updates	Data transfer rate: ~5 Mbps, accuracy: 90%–95%	[48,53,89]

capture license plate images for fee calculation, and enhance security. Their system includes motorized camera heads and zoom lenses for improved image capture, demonstrating the integration of advanced imaging technologies in SPS [96]. Ruili et al. (2019) developed an SPS that integrates AI image processing with ultrasonic sensors for occupancy detection and cameras for license plate recognition. This system not only facilitates accurate billing and boosts security but also recommends parking spots based on cost and proximity, showcasing the benefits of combining multiple technologies for comprehensive parking management [97]. Safi et al. (2017) utilized IoT technologies to create a cloud-based SPS featuring a parking reservation service. It uses RFID tags to restrict access to authorized users, employs license plate recognition for security, and includes functionality to detect and deny entry to oversized or overweight vehicles, ensuring proper use of parking facilities [53].

4.3. Vehicular Ad Hoc network (VANET)-based SPS

VANETs enable dynamic and real-time communication between vehicles and parking infrastructure. Lu et al. (2009) discussed a VANET-based SPS that enhances roadside parking with dynamic guidance systems and advanced security features to prevent theft. This system distributes parking information across expansive lots, streamlining the management process and improving user experience by providing real-time updates [98]. Sauras-Perez et al. (2014) introduced a smart parking solution with onboard units in vehicles to identify and secure preferred parking spots. Their system also offers value-added services that help offset operational costs and open new business opportunities, highlighting the economic benefits of VANET-based SPS [99]. Geng & Cassandras (2012) developed an SPS designed to streamline the allocation and booking of parking spaces in urban environments. Their system takes into account various factors such as parking fees and the proximity to the driver's final destination, offering features like reservation guarantees. It fosters communication between the parking infrastructure and the vehicles, thereby improving parking lot detection and overall efficiency [89].

4.4. Wired network-based SPS

Wired networks continue to play a role in certain SPS applications, particularly where reliable and secure data transmission is crucial. Sai Pavan et al. (2013) contributed to SPS innovations with a design that employs loop detectors as sensory instruments to monitor vehicle presence and provide a live count of occupancy on an LCD display. The system's flexibility and effectiveness are notable in its application across various parking environments, including both covered and open-air parking spaces [100]. This study showcases the enduring relevance of wired networks in providing robust and accurate parking data.

4.5. Other SPS

Various other innovative SPS approaches have been developed, utilizing IoT, cloud platforms, and advanced data processing techniques. Khanna & Anand (2016) introduced an IoT-based cloud platform with a smartphone application to monitor parking space availability in real time. This system enhances user convenience by providing up-to-date parking information directly to their mobile devices [101]. Atif et al. (2016) proposed an SPS leveraging IoT and multi-agent systems to offer dynamic pricing for parking space providers and real-time availability information for drivers. This approach demonstrates the potential for IoT and AI to optimize parking management and pricing strategies [102]. Chatzigiannakis et al. (2016) designed a privacy-preserving SPS using elliptic curve cryptography for devices with limited computational power, enhancing user privacy without compromising sensitive data [57]. Mukadam & Logeswaran (2020) created an SPS using IoT and sensors to detect smart vehicles, incorporating a compact computing device, sensors, and a smartphone application for occupancy monitoring. This integration demonstrates the efficiency of combining IoT and mobile technology for effective parking management [103]. Luque-Vega et al. (2020) utilized IoT to help drivers find nearby parking spaces by offering real-time traffic information and optimal routing suggestions. They also employed edge computing to improve data processing efficiency, highlighting the role of advanced computing techniques in modern SPS [52]. Mathur et al. (2010) used GPS and ultrasonic rangefinders to identify parking spots, addressing data-sharing challenges that led to congestion. Their study emphasizes the importance of reliable data transmission in preventing parking-related congestion [104]. Lan et al. (2014) introduced a crowdsourced real-time parking detection system using smartphone sensors and map-matching algorithms for accuracy in indoor settings. This approach leverages the power of crowdsourcing and mobile technology to enhance parking detection [105]. Salpietro et al. (2015) investigated an SPS based on a mobile app utilizing Bluetooth and embedded sensors for parking initiation, with remote server connections via Wi-Fi for data sharing. This system highlights the potential for seamless integration of mobile and network technologies in SPS [106].

5. Discussion

The rapid urbanization and increasing vehicle ownership worldwide have led to significant challenges in urban mobility and parking management. Traditional parking systems are often inefficient, resulting in traffic congestion, increased emissions, and frustration for drivers. Studies estimate that up to 30% of urban traffic congestion is caused by drivers searching for parking spaces, leading to delays, increased fuel consumption, and higher levels of air pollution [107]. This congestion not only impacts the environment but also results in economic losses due to wasted time and fuel, affecting businesses and local economies [108]. Moreover, inefficient parking systems contribute to user frustration, deterring people from visiting certain areas and impacting local businesses and services [109].

To address these challenges, there is a growing need for advanced SPS that leverage modern technologies to provide efficient, user-friendly, and environmentally sustainable parking solutions. Future SPS development is motivated by several key factors. As cities become smarter, there is a need for integrated parking solutions that can communicate with other urban systems such as traffic management, public transportation, and energy grids [110]. Future SPS will need to be highly interoperable and capable of seamless integration with smart city infrastructures. The rapid advancement of technologies such as IoT, AI, and blockchain offers new opportunities for developing more sophisticated and efficient SPS. These technologies can enhance real-time data processing, improve security, and provide predictive analytics for better parking management [111, 112]. Many cities are setting ambitious sustainability goals to reduce their carbon footprint and promote greener urban environments. Efficient parking management can significantly reduce vehicle emissions and optimize the use of urban space, contributing to these sustainability goals [113]. There is an increasing demand for more convenient and user-friendly parking solutions. Drivers expect real-time information, easy payment options, and seamless navigation to available spaces. Future SPS will need to prioritize user experience, offering intuitive interfaces and reliable services [114]. Efficient parking systems can generate significant economic benefits by reducing operational costs and increasing revenue from parking fees. Future SPS will need to provide cost-effective solutions that maximize the use of available resources [115,116].

5.1. Challenges and innovations

The landscape of sensor technology in SPS is marked by diversity and innovation and serves as a foundational element in the evolution of urban parking solutions. Researchers are actively exploring the strategic selection and deployment of sensors, propelling SPS towards more

efficient, reliable, and user-centric parking management. These technologies are not without their challenges such as environmental susceptibility and the limitations of line-of-sight requirements for camera-based systems [26,117], however, but innovations like radar and microwave sensors, and sensor fusion techniques are emerging as solutions to these hurdles, enhancing vehicle detection accuracy and system robustness across various conditions.

Also critical is the evolution of communication networks within SPS as it transitions from traditional wired infrastructures to advanced wireless technologies to boost system connectivity and scalability. This shift brings its own set of challenges, including signal interference and the imperative for secure, energy-efficient communication [54,94,118]. These issues are being addressed by the adoption of low-power wireless standards and secure communication protocols to ensure that SPS networks are efficient, secure, and capable of safeguarding the user's data and the system's integrity.

The innovative integration of computational models and analytics refine parking operations and enhance user experiences on the data management front, and the application of machine learning and deep learning is making parking systems more intelligent and adaptive, though challenges in data accuracy, volume management, and privacy remain [70,74,119]. Sophisticated algorithms and robust data-handling mechanisms are critical for overcoming these challenges and ensuring precise occupancy predictions and the secure handling of user data.

The transition towards smartphone-based SPS interfaces reflects a significant shift towards meeting modern user expectations for convenience and functionality. These applications, which allow users to locate, reserve, and pay for parking seamlessly, are transforming the parking experience [52]; however, the digital divide and the need for inclusive design present challenges in optimizing them for all users. Balancing functionality with simplicity and ensuring the accessibility of SPS applications is crucial for making smart parking solutions universally adoptable and effective.

In summary, the journey of SPS from sensor technology to user interfaces is one of continuous innovation and adaptation that addresses both technological challenges and user needs. Each step forward in sensor deployment, network communication, data management, and interface design bring SPS closer to achieving the goal of streamlined, user-friendly urban parking solutions, albeit with an ongoing need to navigate the complexities of technology integration and user inclusivity. Table 6 presents the comprehensive comparison of different SPS.

5.2. Emerging technologies and future directions

The future of SPS is poised for transformative innovations, particularly in the realm of sensor technology. The integration of artificial intelligence (AI) and machine learning with sensor data promises to revolutionize how SPS interprets and responds to parking dynamics. AI algorithms have the potential to analyze complex patterns from sensor data, predicting parking availability, optimizing space allocation, and guiding drivers to available spots in real-time. Advancements in sensor technology, such as the exploration of LiDAR sensors originally used in autonomous vehicles, could offer high-resolution, three-dimensional data about parking environments. This would improve vehicle detection accuracy, as well as significantly enhance the overall efficiency of SPS by making it more energy efficient, resilient to environmental factors, and capable of providing richer data sets.

The communication networks within SPS are also on the brink of significant enhancement, driven by the emergence of technologies such as 5G, IoT protocols, and blockchain. The introduction of 5G could dramatically improve the speed, capacity, and reliability of wireless communication, facilitating a higher density of sensor nodes and enabling functionalities like autonomous vehicle integration. IoT protocols could streamline interoperability between diverse devices and systems within SPS to foster a more cohesive urban mobility ecosystem. Meanwhile, blockchain technology holds promise for enhancing data security and transparency in parking transactions, ensuring the integrity and trustworthiness of the data exchanged within the system.

In the area of data management, SPS stands to benefit greatly from the integration of big data analytics, edge computing, and blockchain. Big data analytics can provide deeper insights into parking patterns and user behavior, enabling the development of more user-centric and efficient parking solutions. Edge computing addresses the challenge of managing large volumes of data by processing it closer to the source, which reduces latency and improves system responsiveness. The application of blockchain could further secure data transactions within SPS, thus maintaining the integrity of parking data and fostering user trust.

The user interfaces of SPS are also set for a revolution with the integration of augmented reality (AR), voice-activated controls, and AI-driven predictive analytics. AR could offer an immersive and intuitive experience by overlaying parking availability information directly onto drivers' real-world views. Voice-activated controls promise hands-free operation that will enhance safety and convenience. AI-driven predictive analytics have the potential to personalize the parking experience by learning individual habits and preferences and suggesting optimal parking solutions tailored to each user, thereby significantly enhancing user satisfaction and system efficiency. User feedback from pilot implementations of these technologies has shown significant improvements in satisfaction and ease of use, with many users appreciating the seamless and intuitive nature of these advanced interfaces [120].

SPS integration with broader smart city infrastructures can create synergies that enhance urban mobility. For instance, integrating SPS with traffic management systems can optimize traffic flow and reduce congestion [121]. Seamless communication with public transportation systems can provide drivers with real-time information on alternative transport options, promoting multimodal transport solutions [122]. Moreover, integration with energy grids can facilitate the efficient use of EVs and the implementation of dynamic pricing strategies based on real-time demand, further enhancing sustainability and efficiency [123,124].

These innovations address current challenges while opening new avenues for research and development. The exploration of sustainable energy solutions for sensor networks; the adoption of holistic approaches that consider social, economic, and environmental impacts; and the

Table 6
Comparison of SPS.

SPS	Advantages	Disadvantages	Key technologies	Quantitative metrics
Wired network-based SPS	High reliability, secure transmission	High installation cost, less flexibility	Loop detectors, fiber optic cables	Latency: <1 ms, Installation cost: High
WSN based SPS	Real-time tracking, efficient management	Limited coverage, interference issues	Infrared, ultrasonic sensors	Accuracy: 85%–95%, Communication range: ~100 m
Image processing-based SPS	Detailed monitoring, automated recognition	Affected by lighting conditions	Camera sensors, AI algorithms	Accuracy: ~90%, Processing time: ~5–10 ms
VANET based SPS	Dynamic, real-time communication	High implementation cost, complexity	Onboard units, communication Networks	Communication latency: <50 ms
IoT-based SPS	Flexible, scalable, user-friendly	Privacy concerns, high data bandwidth	IoT sensors, cloud Platforms	Data transfer rate: Up to 10 Gbps (5G)

integration of emerging technologies like blockchain for secure transactions represent potential areas for future innovation. As these technologies mature and integrate, SPS could evolve into more sustainable, efficient, and user-friendly systems that significantly impact urban mobility and the broader transportation ecosystem.

6. Conclusion

Smart parking systems are continually evolving because of the diverse approaches and technological advancements in the field. The systems today are a far cry from the parking meters of yesterday, as they can accurately detect parking occupancy in real time, and, due to the integration of IoT technology, make the information easily accessible to drivers via smartphone applications. This paper explores the complex landscape of SPS, focusing on the integration of advanced technologies. The in-depth analysis of 124 studies, covering methodologies, sensor technologies, communication frameworks, and predictive models, aims to provide a comprehensive overview of the field, highlight the innovative strides within SPS, and identify areas primed for future exploration and improvement.

The current review brought to light novel sensor technologies that enhance vehicle detection accuracy and environmental adaptability. Therefore, we propose that rather than relying on more conventional ultrasonic or infrared sensors, an integrated sensor approach be taken that combines the precision of radar sensors with the wide-area coverage of camera sensors, augmented by AI-based image processing algorithms. This hybrid sensor model leverages the strengths of each sensor type, ensures reliable vehicle detection across diverse conditions, and reduces false positives that plague single-sensor systems. In the realm of communication networks, we introduce an innovative mesh network architecture that surpasses the traditional wired and wireless frameworks in scalability and reliability. It is characterized by its self-healing and dynamic routing capabilities and ensures uninterrupted data flow even in the event of individual node failures, making it exceptionally suited for the complex and dynamic environment of urban parking. Our discussion on data management systems reveals a pioneering application of blockchain technology to ensure data integrity and security in SPS by addressing the pressing concerns of privacy and data tampering, thus ensuring that parking transactions and vehicle data are immutable and transparent. This approach not only enhances security but also fosters user trust in SPS. Our review proposes a novel adaptive user interface design that personalizes the user experience based on historical usage patterns and contextual information. This AI-driven interface anticipates user needs and simplifies the parking process, making it more intuitive and user-friendly, especially in high-stress urban parking scenarios.

In conclusion, this literature review transcends a mere synthesis of existing knowledge by injecting novel ideas and innovative solutions into the discourse on smart parking systems. Future research on SPS can explore advanced sensing technologies and methods that will optimize data transmission and intelligent data analytics, enhance user experience, and integrate the systems with those of smart transportation systems. Advancements in sensing technologies will improve accuracy and scalability, intelligent data analytics will optimize resource allocation and predict parking demand patterns, and integration with smart transportation systems will enhance mobility solutions that will be energy-efficient and contribute to sustainability. User-centric designs and consideration of human factors will ensure usability and privacy. By focusing on these areas, future studies can contribute to the development of smarter and more sustainable SPS in the context of evolving smart cities.

CRediT authorship contribution statement

Sai Sneha Channamallu: Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Sharareh Kermanshachi:** Writing – review & editing, Conceptualization. **Jay Michael Rosenberger:** Writing – review & editing, Conceptualization. **Apurva Pamidimukkala:** Writing – review & editing, Conceptualization.

Data availability statement

The data and materials used to support the findings of this study are available from the corresponding author upon reasonable request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could influence the work reported in this paper.

Acknowledgements

We gratefully acknowledge the support and generosity of the North Central Texas Council of Governments (NCTCOG), without which the present study could not have been completed.

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